

EE2080: Random Processes

Review: Probability and Random Variables

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Probability Space

Probability Theory deals with modeling

- **Random experiments:** Results are not deterministic, & their
- **Outcomes** : the possible results of the experiment.

A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- **Sample space:** the set of all possible outcomes, denoted Ω
- **Sigma-algebra** of subsets of Ω - A collection $\mathcal{F} \subseteq 2^\Omega$ s.t.

E1. $\Omega \in \mathcal{F}$

E2. Closed under complements: $A \in \mathcal{F} \implies A^c \in \mathcal{F}$

E3. Closed under countable unions:

$$A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

Probability Space: Continued

- The pair $(\Omega, \mathcal{F})$: **Measurable Space**
- **Probability measure** $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$
 - P1. Non-negativity: For any $A \in \mathcal{F} : \mathbb{P}(A) \geq 0$
 - P2. Normalization: $\mathbb{P}(\Omega) = 1$
 - P3. Countable additivity: If A_i are disjoint, $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$

Example

Consider the experiment of rolling a fair die.

- Sample Space $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Sigma-algebra:
 - $\mathcal{F}_1 = \{\emptyset, \Omega\}$
 - $\mathcal{F}_2 = 2^\Omega$
 - $\mathcal{F}_3 = \{\emptyset, \{1, 2\}, \{3, 4, 5, 6\}, \Omega\}$
 - $\mathcal{F}_4 =$
 $\{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{3, 4, 5, 6\}, \{1, 3, 4, 5, 6\}, \{2, 3, 4, 5, 6\}, \Omega\}$
- Consider $\mathcal{F}_3 = \{\emptyset, \{1, 2\}, \{3, 4, 5, 6\}, \Omega\}$
 - $\mathbb{P}_1 = \{0, \frac{1}{3}, \frac{2}{3}, 1\}$
 - $\mathbb{P}_2 = \{0, \frac{1}{6}, \frac{5}{6}, 1\}$

Borel Sigma-Algebra

- $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all open intervals of $\mathbb{R}$
- Includes all open/closed/half-open intervals, countable unions, intersections, complements
- It is the smallest σ -algebra containing sets of the form $(-\infty, x], \forall x \in \mathbb{R}$.

Remarks:

- Ω is called a sure event
- Any $A \in \mathcal{F}$ s.t. $\mathbb{P}(A) = 1$ is an almost-sure event
- $\mathbb{P}(A) = 1 \not\Rightarrow A = \Omega$
- $\mathbb{P}(A) = 0 \not\Rightarrow A = \emptyset$

Conditional Probability

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Given an event $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the **conditional probability** of an event $A \in \mathcal{F}$, is $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$

- Chain/Multiplication Rule:**

$$\mathbb{P}(\cap_{i=1}^n A_i) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_1 \cap A_2) \cdots \mathbb{P}(A_n|\cap_{i=1}^{n-1} A_i)$$

- Total Probability Theorem:** Let $A_1, A_2, \dots, A_n$ be disjoint events that partition the sample space. Assume that $\mathbb{P}(A_i) > 0$. Thus, for any $B \in \mathcal{F}$, we have

$$\mathbb{P}(B) = \sum_{i=1}^n \mathbb{P}(A_i)\mathbb{P}(B|A_i)$$

- Bayes' Rule:** $\mathbb{P}(A_i|B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B|A_i)}{\sum_{i=1}^n \mathbb{P}(A_i)\mathbb{P}(B|A_i)}$

Independence

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Events $A, B \in \mathcal{F}$ are **independent** if
 - $\mathbb{P}(A|B) = \mathbb{P}(A)$ (can be used if $\mathbb{P}(B) > 0$) (OR)
 - $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ (can be used even when $\mathbb{P}(B) = 0$)
- Given an event $C \in \mathcal{F}$ with $\mathbb{P}(C) > 0$, A and B are **conditionally independent** if $\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C)$
- Independence does not imply conditional independence and vice-versa.

Fix $(\Omega, \mathcal{F})$.

A random variable $X : \Omega \rightarrow \mathbb{R}$ is a **measurable function**, i.e.,

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F}$$

Equivalently, a mapping $X : \Omega \rightarrow \mathbb{R}$ is a random variable if

$$\forall x \in \mathbb{R}, \quad X^{-1}((-\infty, x]) = \{\omega : X(\omega) \leq x\} \in \mathcal{F}$$

Example: Not a Random Variable

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \{\emptyset, \{1, 2, 3\}, \{4, 5, 6\}, \Omega\}$.

Define $X(\omega)$ as

$$X(\omega) = \begin{cases} 1, & \omega \in \{1, 2, 3\} \\ 2, & \omega \in \{4\} \\ 3, & \omega \in \{5, 6\} \end{cases}$$

Then X is not a random variable since

$$X^{-1}((-\infty, 2.5]) = \{1, 2, 3, 4\} \notin \mathcal{F}.$$

Question: For the given $\mathcal{F}$, which functions are RVs?

CDF of a Random Variable

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Given a random variable $X : \Omega \rightarrow \mathbb{R}$, its **cumulative distribution function** (CDF), $F_X : \mathbb{R} \rightarrow [0, 1]$, is defined as

$$F_X(x) = \mathbb{P}((-\infty, x]), \quad x \in \mathbb{R}$$

Properties of CDF:

1. $\lim_{x \rightarrow -\infty} F_X(x) = 0, \lim_{x \rightarrow \infty} F_X(x) = 1$
2. **Monotonicity** If $a \leq b$, then $F_X(a) \leq F_X(b)$
3. **Right-Continuity** F_X is right-continuous, i.e., for all $x \in \mathbb{R}$, $\lim_{\epsilon \downarrow 0} F_X(x + \epsilon) = F_X(x)$.

Discrete vs Continuous Random Variables

Discrete RV:

- Takes countable values
- Described by **Probability Mass Function (PMF)**:

$$p_X(x) = \mathbb{P}(X = x)$$

- Common Distributions: Bernoulli, Binomial, Geometric, Uniform, Poisson

Continuous RV:

- Takes values in intervals of $\mathbb{R}$
- Described by **Probability Density Function (PDF)**:

$$f_X(x) \text{ s.t. } \mathbb{P}_X(a \leq X \leq b) = \int_a^b f_X(x) dx$$

- Common Distributions: Uniform, Exponential, Gaussian

PMFs of Common Distributions

- **Bernoulli** $\text{Ber}(p)$: $p_X(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0, \end{cases} \quad x \in \{0, 1\}.$

- **Binomial** $\text{Bin}(n, p)$:

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

- **Geometric** $\text{Geo}(p)$:

$$p_X(k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

- **Discrete Uniform** $\text{Unif}(\{a, \dots, b\})$:

$$p_X(k) = \frac{1}{b - a + 1}, \quad k = a, a + 1, \dots, b.$$

- **Poisson** $\text{Pos}(\lambda)$:

$$p_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

PDFs of Common Continuous Distributions

- **Uniform** $U[a, b]$:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

- **Exponential** $\text{Exp}(\lambda)$:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

- **Gaussian (Normal)** $N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Joint Distribution of Two Random Variables

Let X and Y be RVs defined w.r.t the same $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Joint CDF:** $F_{X,Y}(x,y) = \mathbb{P}(X \leq x, Y \leq y)$
- If X, Y discrete: **Joint PMF:** $p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$
- **Marginal PMFs:** $p_X(x) = \sum_y p_{X,Y}(x,y),$
 $p_Y(y) = \sum_x p_{X,Y}(x,y)$
- If X, Y jointly continuous: **Joint PDF:** $f_{X,Y}(x,y)$
For $B \in \mathcal{B}(\mathbb{R})$, $\mathbb{P}((X, Y) \in B) = \iint_B f_{X,Y}(x,y) dx dy$
- **Marginal PDFs:** $f_X(x) = \int_y f_{X,Y}(x,y) dy,$
 $f_Y(y) = \int_x f_{X,Y}(x,y) dx$

Expectation

- Expectation: $\mathbb{E}[X]$
 - Discrete RV: $\mathbb{E}[X] = \sum x \cdot p_X(x)$
 - Continuous RV: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$
- **Linearity of Expectation:** $\mathbb{E}(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i \mathbb{E}(X_i)$
- **Fundamental Theorem of Expectation:**
 - $\mathbb{E}[g(X)] = \sum g(x) \cdot p_X(x)$
 - $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx$
- **Tail Probability Formula** for Expectation of a Continuous RV:

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X > x) dx - \int_0^{\infty} \mathbb{P}(X < -x) dx$$

Conditioning and Independence

Let X and Y be RVs defined w.r.t the same $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Conditional PMF:** $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$ for some y s.t $p_Y(y) > 0$
- **Conditional Expectation:** $\mathbb{E}(X|Y = y) = \sum_x x \cdot p_{X|Y}(x|y)$
- **Total Expectation theorem:**
$$\mathbb{E}(X) = \sum_y \mathbb{E}(X|Y = y) \cdot p_Y(y)$$
- **Law of Iterated Expectations:** $\mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(X)$
- **Independence:** $p_{X,Y}(x,y) = p_X(x)p_Y(y), \forall (x,y)$
- **Conditional PDF:** $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ for any y s.t $f_Y(y) > 0$
- If $X \perp Y$, $\mathbb{E}[XY] = \mathbb{E}(X)\mathbb{E}(Y)$

Moments

Consider a RV X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

- n -th Moment: $\mathbb{E}[X^n]$
- n -th Central Moment: $\mathbb{E}[(X - \mathbb{E}[X])^n]$
- The second central moment is called the **variance**.
- **Variance:** $\boxed{\text{var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]}$
- Standard Deviation of X : $\sqrt{\text{var}(X)}$
- For RVs X, Y defined on $(\Omega, \mathcal{F}, \mathbb{P})$,
 - Conditional Variance: $\text{var}(X|Y) = \mathbb{E}[(X - \mathbb{E}(X|Y))^2|Y]$
 - **Law of Total Variance:**
$$\text{var}(X) = \mathbb{E}[\text{var}(X|Y)] + \text{var}(\mathbb{E}(X|Y))$$
 - If $X \perp Y$, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

Covariance and Correlation

Let X and Y be RVs defined w.r.t the same $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Covariance:** $\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))]$
- Alternatively: $\text{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$
- $\text{cov}(X, Y) = 0 \Rightarrow X$ and Y are uncorrelated.
- **Independence $\Rightarrow$ Uncorrelatedness**, converse not true, in general
- **Correlation Coefficient:** $\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$
- $-1 \leq \rho \leq 1$ (Proof by Cauchy-Schwarz Inequality)

Moment Generating Functions

For a RV X associated with the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Moment Generating Function,**

$$M_X(s) = \mathbb{E}[e^{sX}] = \begin{cases} \sum_x e^{sx} p_X(x), & X \text{ discrete} \\ \int_x e^{sx} f_X(x) dx, & X \text{ continuous} \end{cases}$$

- Associated **Region of Convergence** (ROC):

$$\{s : |M_X(s)| < \infty\}$$

- MGF of a continuous RV: Sign reversal of bilateral Laplace Transform
- Moment Generating Properties:

$$M_X(0) = 1, \quad \frac{d}{ds} M_X(s)|_{s=0} = \mathbb{E}(X), \quad \frac{d^n}{ds^n} M_X(s)|_{s=0} = \mathbb{E}(X^n)$$

- **Linear function:** If $Y = aX + b$, then $M_Y(s) = e^{sb} M_X(as)$

Sum of a Random Number of Independent RVs

- Consider $X_1, X_2, \dots$ i.i.d RVs with mean $\mathbb{E}(X)$ and variance $\text{var}(X)$
- Another RV N that takes non-negative values
- Let $Y = X_1 + X_2 + \dots + X_N$
 - $\mathbb{E}(Y) = \mathbb{E}(N)\mathbb{E}(X)$
 - $\text{var}(Y) = \mathbb{E}(N)\text{var}(X) + (\mathbb{E}(X))^2\text{var}(N)$
 - $M_Y(s) = M_N(\log(M_X(s)))$

Concentration Inequalities

- Useful for bounding tail probabilities
- **Markov Inequality:** For a RV $X \geq 0$ with $\mathbb{E}(X) < \infty$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0$$

- **Chebyshev Inequality:** For a RV X with mean μ and variance σ^2 ,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

- **Chernoff Bound:** Exponentially decaying bound

$$P(X \geq a) \leq \inf_{\substack{s \geq 0 \\ s \in \text{ROC}}} \frac{\mathbb{E}[e^{sX}]}{e^{sa}}, \quad a > 0$$

Convergence of a Sequence of Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

$\{X_n\}_{n=1}^{\infty}$ be a **sequence of RVs** defined w.r.t $\mathcal{F}$.

Let X be another RV, also defined w.r.t $\mathcal{F}$.

- **Point-wise Convergence:**

$$X_n \xrightarrow{\text{pointwise}} X \iff \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega), \forall \omega \in \Omega$$

- **Almost-sure Convergence:**

$$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}(\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1$$

- **Convergence in Probability:**

$$X_n \xrightarrow{P} X \iff \forall \epsilon > 0, \lim_{n \rightarrow \infty} \mathbb{P}(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = 0$$

Convergence: Continued....

- **Mean-Square Convergence:**

$$X_n \xrightarrow{m.s.} X \text{ if } \mathbb{E}[X_n^2] < \infty, \forall n, \text{ and } \lim_{n \rightarrow \infty} \mathbb{E}[(X_n - X)^2] = 0$$

- **Convergence in Distribution:** A sequence of RVs $\{X_n\}_{n=1}^{\infty}$ converges to a RV X in distribution,

$$X_n \xrightarrow{d} X \iff F_{X_n}(x) \rightarrow F_X(x) \text{ at all continuity points } x \text{ of } F_X$$

- Implications of various forms of convergence

$$X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X$$

$$X_n \xrightarrow{m.s.} X \Rightarrow X_n \xrightarrow{p} X$$

If $X_n \xrightarrow{d} c$, where c is a constant, then $X_n \xrightarrow{p} c$

Limit Theorems

Let $X_1, X_2, \dots$ i.i.d. with mean μ , variance σ^2

- **Weak Law of Large Numbers (WLLN):**

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

- **Strong Law of Large Numbers (SLLN):**

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu$$

- **Central Limit Theorem (CLT):**

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

1. D. P. Bertsekas, J. N. Tsitsiklis, "Introduction to Probability", 2/e, Athena Scientific.
2. Bruce Hajek, "Random Processes For Engineers", Cambridge University Press, 2013.